

Influencing the Synapse

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

Information text

At the chemical synapse, information is transmitted by the release of neurotransmitters from synaptic vesicles of the presynaptic terminal into the synaptic cleft. The transmitters bind to receptors of the postsynaptic membrane according to the lock-and-key principle and thereby change the opening probability of specific ion channels. This produces postsynaptic potentials that influence the excitability of the downstream cell.

Synaptic transmission can be specifically altered by different substances. Agonists imitate the effect of the body's own transmitters, antagonists block receptors, and other substances inhibit enzymatic breakdown or reuptake of neurotransmitters. As a result, signal transmission is strengthened, weakened or prolonged in time. The chemical synapse is therefore a central target for neurotoxins, medicines and addictive substances. Even small changes at the synaptic level can have considerable effects on information processing, behaviour and motor activity.

Task 1: Compare sites of action

Choose two substances from the "Influencing the Synapse" mode. Export two suitable states and insert them below.

Substance 1: _____ Substance 2: _____

Insert here

Insert here

Task 2: Effect on the postsynaptic reaction

Describe how the selected active substances affect the synapse. Explain whether signal transmission is strengthened, weakened, prolonged or prevented.

Task 3: Consequences for the organism

In the table, add whether the effect of the different substances leads to flaccid paralysis or rigid paralysis. Use the modes of action in the simulation.

Active substance	Effect
Curare	
Botulinumtoxin (<i>Botox</i>)	
Sarin	
Nikotin	
α -Latrotoxin	
Conotoxin	